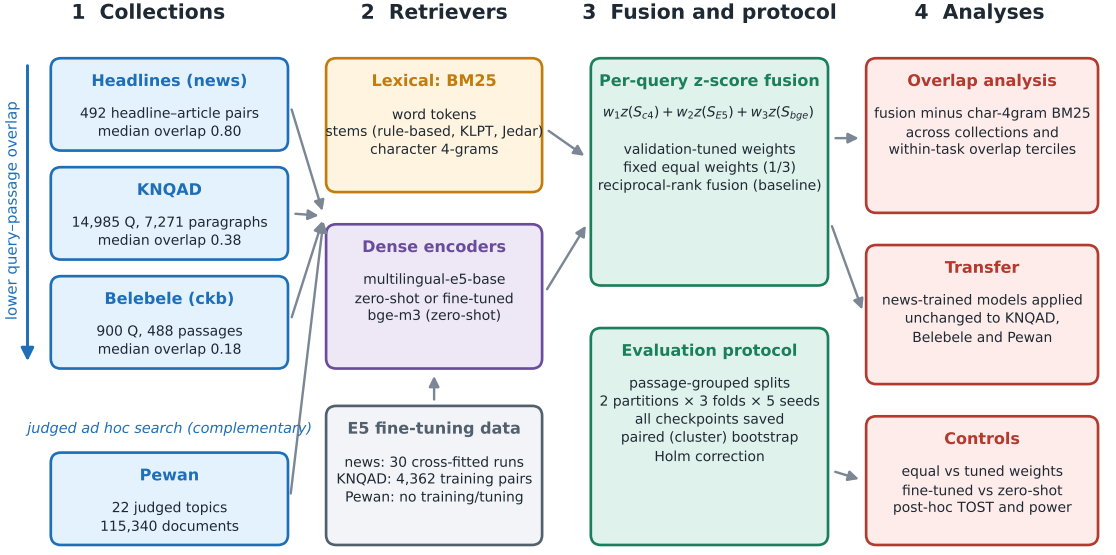
1 Collections
2 Retrievers
3 Fusion and protocol
4 Analyses
lower query–passage overlap
Headlines (news)
492 headline–article pairs
median overlap 0.80
KNQAD
14,985 Q, 7,271 paragraphs
median overlap 0.38
Belebele (ckb)
900 Q, 488 passages
median overlap 0.18
judged ad hoc search (complementary)
Pewan
22 judged topics
115,340 documents
Lexical: BM25
word tokens
stems (rule-based, KLPT, Jedar)
character 4-grams
Dense encoders
multilingual-e5-base
zero-shot or fine-tuned
bge-m3 (zero-shot)
E5 fine-tuning data
news: 30 cross-fitted runs
KNQAD: 4,362 training pairs
Pewan: no training/tuning
Per-query z-score fusion
$w_1 z(S_{c4}) + w_2 z(S_{E5}) + w_3 z(S_{bge})$
validation-tuned weights
fixed equal weights (1/3)
reciprocal-rank fusion (baseline)
Evaluation protocol
passage-grouped splits
2 partitions × 3 folds × 5 seeds
all checkpoints saved
paired (cluster) bootstrap
Holm correction
Overlap analysis
fusion minus char-4gram BM25
across collections and
within-task overlap terciles
Transfer
news-trained models applied
unchanged to KNQAD,
Belebele and Pewan
Controls
equal vs tuned weights
fine-tuned vs zero-shot
post-hoc TOST and power